A3-PS003-TC141

Federated Traffic-Flow Prediction with Communication-Cost Evaluation

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Problem Description

Abstract. We propose a simulator comparing centralized and federated traffic-flow prediction across virtual intersections. Clients train locally; a server combines model updates using Federated Averaging (FedAvg).
We will compare prediction error, communication cost, and convergence under IID and non-IID data using the UCI Traffic Flow Forecasting dataset [2]. The study uses 3–5 clients, chronological train/test splitting, and three random seeds. Results will be reported after experiments.

Problem being addressed

- Centralized training pools data and may require raw-data transfer.
- Intersections can have different traffic patterns (non-IID data).
- Federated learning keeps training data local but exchanges model updates.
- We will measure the trade-off using MAE, RMSE, bytes per round, and convergence.

Scope: UCI traffic data; 3–5 clients; one compact model; local, centralized, and FedAvg comparisons; IID/non-IID partitions; three seeds. No live cameras or formal privacy guarantees.

Background

Traffic forecasting estimates future conditions from historical measurements. Centralized training pools data from multiple locations; federated learning trains at each client and exchanges model updates instead of routinely pooling raw examples.

Federated Averaging (FedAvg)

Broadcast global model → train locally → send updates → aggregate → repeat.

Client-data settings

- **IID:** clients have broadly similar data distributions.
- **Non-IID:** clients have different traffic volumes or rush-hour patterns.

We will measure prediction error and convergence. Federated learning limits routine raw-data sharing but does not guarantee formal privacy.

Literature & RFC Review

FedAvg provides the server-aggregation method for decentralized training [1]. Traffic-flow forecasting literature motivates spatial and temporal features [3], while federated-optimization research highlights non-IID client data as a challenge [4].

Evaluation focus

- Compare local-only, centralized, and FedAvg models.
- Test IID and non-IID client partitions.
- Measure MAE, RMSE, transmitted bytes, and convergence.

Project focus

A reproducible software simulation using one public dataset and 3–5 virtual clients.

Standards: no mandatory RFC is specified in PS-003.

Proposed Solution

A software-only simulator will use the UCI Traffic Flow Forecasting dataset (36 sensor locations, 15-minute intervals) [2]. We will validate the file structure, then split data chronologically across 3–5 virtual clients; the test period stays held out.

Main workflow

1. **Prepare data:** verify source/licence, target, and chronological split.
2. **Simulate clients:** create IID and non-IID training partitions.
3. **Build baselines:** train centralized and local-only models.
4. **Federate:** train locally, aggregate with FedAvg, broadcast the global model.
5. **Evaluate:** measure MAE, RMSE, bytes per round, and convergence.
6. **Repeat:** run experiments with three random seeds.

We will start with a compact forecasting model and confirm the dataset loader before training. No physical sensors, live cameras, or city-wide deployment are required.

Solution Optimization

We will compare prediction quality and communication cost across local-only, centralized, and federated training. Federated runs will use IID and non-IID partitions.

Planned comparison

Approach	Training arrangement	Purpose
Local-only	Each client trains on its own data	Non-collaborative baseline.
Centralized	One model uses pooled training data	Centralized reference.
Federated IID	Similar client distributions; FedAvg	Test IID federated training.
Federated non-IID	Non-identical client data; FedAvg	Test client heterogeneity.

Metrics

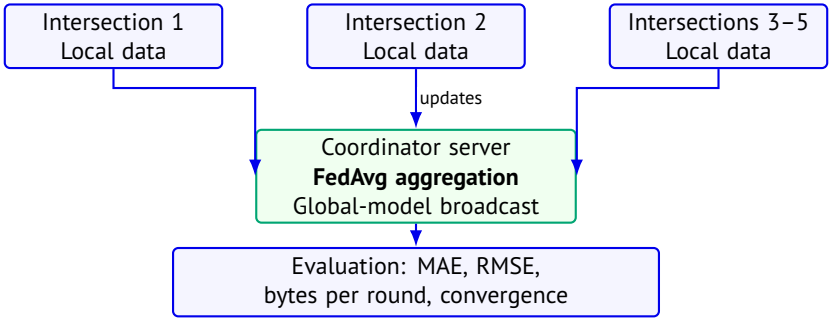
- **MAE/RMSE:** prediction error (lower is better).
- **Bytes per round:** serialized model-update traffic.
- **Convergence:** performance tracked across rounds.

Use chronological splitting and the same held-out test period. Report improvements only if measured.

Architecture Design (UML Diagram)

Solution Architecture & Design

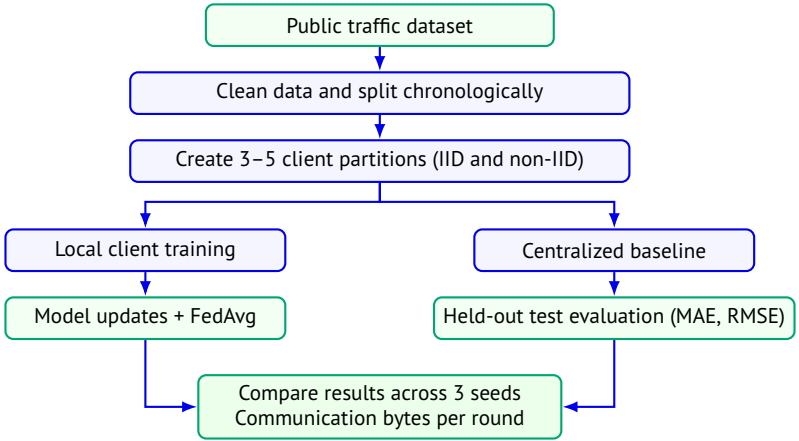
Local training, server aggregation, and global-model broadcast.



Local training, FedAvg aggregation, global-model broadcast.

Data Flow Diagram

Prepare data → train clients → compare results.



Held-out test data; communication measured each round.

Deployment Timeline

Development milestones; dates will be confirmed by the team.

1. **Validate data:** inspect UCI file, licence, target, and time split.
2. **Build baselines:** preprocess data; train centralized/local models.
3. **Simulate clients:** create 3–5 clients and IID/non-IID partitions.
4. **Implement FedAvg:** local training, aggregation, and broadcast.
5. **Measure:** MAE, RMSE, bytes per round, and convergence.
6. **Repeat:** run three random seeds and compare approaches.
7. **Demo and report:** validate, present results, document limitations, and submit files.

Final checks

- No future observations enter training.
- Comparisons use the same test period.
- Byte measurements and all seed results are retained.

Bibliography

- H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, “Communication-Efficient Learning of Deep Networks from Decentralized Data,” in *Proceedings of AISTATS*, 2017, pp. 1273–1282. AISTATS paper
- L. Zhao, “Traffic Flow Forecasting,” UCI Machine Learning Repository, 2019. DOI: 10.24432/C57897. CC BY 4.0. Dataset page.
- L. Zhao, O. Gkountouna, and D. Pfoser, “Spatial Auto-regressive Dependency Interpretable Learning Based on Spatial Topological Constraints,” *ACM Transactions on Spatial Algorithms and Systems*, vol. 5, no. 3, Article 19, 2019. DOI: 10.1145/3339823.
- T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, “Federated Optimization in Heterogeneous Networks,” in *Proceedings of Machine Learning and Systems*, vol. 2, 2020. MLSys paper

Dataset candidate selected for the proposal; validate the downloaded file and cite the source in all derivative work.

Thank You for Your Attention!

Team RESONANCE

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